Repetition, Similarity, Relevance

Toward an Integrated Retrieved Context Account of Text Comprehension and Free Recall

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Abstract

I propose experiments to specify and evaluate competing explanations of text comprehension and memory in terms of retrieved context theory (RCT). RCT explains recall of word sequences by asserting that 1) studied items are associated with a gradually changing, recency-weighted representation of temporal context, 2) the contexts retrieved from studying (or recalling) items update the current state of context, and 3) the current state of context is the proximate cue for each item recall. RCT has been successful in accounting for a wide range of free and serial recall data, explaining key benchmark phenomena, predicting the effects of experimental manipulations, and generalizing to naturalistic research conditions. Comprehension, the process of constructing an integrated mental representation of the meaning of a text, is known to engage many of the same cognitive processes as the free and serial recall tasks, and preliminary evidence suggests that text memory data exhibits many of the same patterns observed in free recall. However, an integrated account of text comprehension and memory in terms of RCT has remained elusive. Here, I argue that the main obstacle to an integrated account is underspecification of how similarity and feature overlap across study materials interact with temporal context to structure the associations learned during study and applied to guide recall. The proposed research will narrow this gap by constructing parallel free recall datasets using stories and word lists as study materials and manipulating relations between study events via item repetition, item similarity, and item co-relevance. Through formal comparison of candidate RCT-based models, the proposed research will clarify key constraints for an integrated account of text comprehension and memory to guide future research and theory development.

About this version

This is an earlier research proposal, preserved in February 2024 and published here in October 2026. It records the original plan to connect repetition, semantic reminding, and narrative memory. Subsequent repetition-memory work challenged the neighbour-contiguity evidence used as a starting point here, sharpening the question of when reinstatement changes learning. The original proposal text is retained below, including its provisional claims and citations. This sketch is separate from the full Vanderbilt thesis proposal submitted in September 2024, which is available on its own site with separate chapter pages and the original 165-page PDF.

Introduction

Models of text comprehension and of serial memory share a common thread: both concern how studying sequences of items come to be associated with one another

Premise

According to Retrieved Context Theory,

1. Memory associates each study event in a list learning experiment with a state of temporal context, a representation that reflects your recent experiences.
2. When you study an item in a list-learning experiment, it reminds you of past experiences with that item. You retrieve its pre-existing contextual associations and integrate that into the current temporal context.

These two points have interesting implications in a list with item repetitions such that participants study some items at two positions, i and j. When you see the item at position j, you reinstate the temporal context from when you saw it at position i. Because of (1), a side effect of this reminder is that context will also reinstate neighbors of the item's first presentation ($i+1$, $i+2$). This information will persist in context for a few study events, so the item's first presentation neighbors get associated with the item's second presentation neighbors. Even though the items at $i+1$ and $i+2$ were never restudied!

Lohnas & Kahana (2014) identified this implication and showed that CMR predicts this neighbor contiguity effect and that the effect shows up in behavioral data. When people later recall an item in $\{i+1, i+2\}$, they transition to recalling items in $\{j+1, j+2\}$ at rates much higher than chance. Even if they never actually recall the item repeated at positions i and j.

This ties into semantic or narrative memory because this analysis helps detect *any* reminding effects in a free recall experiment. Lynn's results suggest that when you see an item for the second time in a list learning experiment, you tend to remember the first time you saw it in a way that shapes the course of recall. We can ask if this happens when you see **two distinct but highly related ideas** in a study sequence.

If I see the word KING at position i and then the word QUEEN at position j in a study list, do I get reminded of my experience of the word KING in a way that shapes how I process the list going forward and later remember it?

If I read a sentence about how dangerous wild animals are at one point in a narrative and then a bear occurs as a character later, do I get reminded of the "wild animals" sentence in a way that shapes how I process the story going forward and later remember it?

I am sure a positive answer to the latter question is already common knowledge in the narrative comprehension literature – they call the phenomenon bridging inference and make a big deal about it. I'd go as far as to say that most model competition in the narrative comprehension literature concerns when and how readers perform bridging inferences to comprehend a text. For example, an ongoing debate in the narrative comprehension literature distinguishes between strategy-based and memory-based accounts of bridging inference (O'Brien & Cook, 2015). Strategy-based models assume that readers actively search the long-term memory representation of the text for information relevant to the contents of working memory. In contrast, memory-based models assume that inactive information related to the current input is reactivated via a passive "resonance mechanism".

By generalizing Lohnas & Kahana's (2014) neighbor contiguity analysis to measure reminding effects driven by semantic similarity instead of just item repetitions, we can obtain a way to relate the problem of bridging inference in the narrative comprehension literature to parallel issues surrounding semantic organization in free recall.

The models examined in Morton & Polyn (2016) don't quite explore this question because they don't allow semantic similarity to influence the encoding process. Instead, the models all suppose that semantic similarity between items only directly affects the recall process. They thus don't predict a "similar items" neighbor contiguity effect that a semantic resonance mechanism would enact.

I think with the right design, though, we can show that such an effect does exist and that a straightforward modification to CMR can account for it. Furthermore, I think we can use the same approach to show that this same model can account for a range of bridging inference phenomena in narrative comprehension. If successful, it would draw a direct line from this core idea in narrative comprehension — bridging inference — to a core idea in the list-learning literature — context reinstatement. And hopefully set the stage for a unified account of both processes.

Experiment 1: Demonstrating the Principle

To investigate whether similarity-driven contextual reinstatement organizes the list learning process, we can start by replicating part of Lynn's 2016 experiment.

Participants were presented with control lists and mixed lists. Each list contained 40 unique positions. In the control lists, 40 unique items were presented just once. In the mixed lists, 6 pairs of repeated items were mixed among 28 once presented items. Across lists, there were an equal number of item pairs repeated at spacings of lag $\in \{0, 1, 2, \ldots, 8\}$, where lag is defined as the number of intervening items between an item's repetitions. 35 subjects saw lists of each type 12 times over 3 sessions.

To analyze the data, Lynn measured the percent of transitions between items that follow the presentation of the same repeated item. If an item was presented at positions i and j, and if one of the items was recalled from either $S_j = \{j+1, j+2\}$ or $S_i = \{i+1, i+2\}$, they then counted the percentage of times that a transition was then made to one of the items in the complementary sets S_i or S_j, respectively.

To measure the percent of transitions expected by chance, over 100 reshuffled datasets, repeated items in control lists were matched to the same serial positions as the repeated items in mixed lists. Then the percentage of transitions was counted.

We can replicate these results to measure similarity-driven contextual reinstatement and add a condition similar to our mixed list condition. In mixed lists, six pairs of repeated items were mixed among 28 once presented items. In our additional condition, we can use six pairs of items with high semantic similarity to one another but low semantic similarity to other list items. We can enforce the remaining 28 items to have comparatively low semantic similarity to all list items. Then to analyze the data, we can measure the percent of transitions between items that follow the presentation of items in the same pairing.

If we observe that this percent of transitions is significantly higher than chance, we can conclude that similarity-driven contextual reinstatement influences list learning and the course of free recall.

If we don't, but under the same conditions can replicate Lynn's results regarding the mixed list condition, we can surmise that similarity-driven contextual reinstatement does not influence list learning.

It may be possible to do a similar demonstration with existing datasets, but I figure a design like this would provide a clearer test of the semantic resonance hypothesis.

Modeling

Let's suppose do find a resonance effect. With data from Experiment 1 and by integrating a similarity-driven contextual reinstatement mechanism into CMR, we can also test the hypothesis through model comparison. We can demonstrate that the mechanism contributes unique predictions about the outcome of our neighbor contiguity analysis in our novel experimental condition. And through likelihood-based model fitting, we can show that these differences either improve or worsen the model's capacity to account for relevant datasets overall.

During list learning, the mechanism would alter how the model calculates contextual input c_i^{IN}. In the classic specification of CMR, $c_i^{IN} = M^{FC} f_i$, where M^{FC} is a matrix that learns associations between orthogonalized item representations and contextual states. In a modified model, we might maintain a parallel memory structure M^{sem} that learns associations between distributed semantic representations of items and contextual states. This way, contextual reinstatement upon processing an item will be graded based on the similarity of the item's semantic representation to semantic representations of other items in memory.

I'm unsure, but these associations may prove easier to implement within an instance-based architecture than a traditional linear associator model. A nonlinear activation scaling mechanism could control the extent to which the model reinstates similar items' contextual representations upon item processing.

Experiment 2: Generalization to Arbitrary Word Lists

A natural follow-up to our results in Experiment 1 is generalizing the experiment to arbitrary word lists. Using benchmark datasets from Morton & Polyn (2016), we can perform a model comparison to evaluate whether the hypothesized mechanism can generalize to improve CMR's account of memory for word lists with less structured similarity relations between studied words.

These results would help clarify the scope of results in Experiment 1, ruling out or confirming that

1. the design of our original experiment influenced subjects to process paired high-similarity words like item repetitions, but that subjects do not do so naturally.
2. similarity-driven contextual reinstatement can influence free recall even without enforced extremes in the semantic similarity between items that I propose we enact in Experiment 1

Experiment 3/4: Generalization to Written Narratives

Finally, we can attempt to generalize our results to memory for written narratives.

Two approaches are possible. On the one hand, we can try to create narratives whose structures mirror those of the control, mixed, and similarity-pairing lists from our Experiment 1. Each narrative can contain a connected sequence of 40 "idea units" or subjects/predicates in single words or phrases. In our "control" narratives, we'll enforce idea units to be unique and have low similarity to one another. In our "mixed" narratives, 6 pairs of repeated idea units can be mixed among 28 once presented idea units. And in our similarity-pairing condition, we can use 6 pairs of idea units with high semantic similarity to one another but a low semantic similarity to other idea units. The remaining 28 idea units will be enforced to have comparatively low semantic similarity to all idea units.

This design might enable a clear demonstration of a similarity-driven contextual reinstatement mechanism in narrative recall but seems to require constructing very unnatural stories since a

typical narrative includes some item repetition. So alternatively, we might use naturally structured narratives identified from previous research into narrative comprehension. Then we might apply the likelihood-based model comparison technique from Morton & Polyn (2016) to evaluate how including a similarity-driven contextual reinstatement mechanism in CMR influences the model's ability to account for narrative recall datasets.

At the same time, we'd also be evaluating CMR itself as an account of narrative recall. We may include implementations of other comprehension models in this analysis.

Discussion

So what do we get from all this trouble? Ideally, a measure of theoretical unification.

Similarity-driven contextual reinstatement is a powerful mechanism wielded in models of narrative comprehension to account for readers' ability to organize otherwise seemingly disjointed text ideas into a coherent discourse representation. It may also be an important way semantic knowledge shapes new episodic memories in general. These proposed experiments suggest a way we might convincingly connect these ideas under a single familiar theoretical framework – retrieved context theory.

Our extension of the neighbor contiguity analysis from Lohnas & Kahana (2016) gives us a way to cleanly demonstrate the behavioral predictions suggested by the mechanism under controlled conditions and independently of formal theoretical commitments. It also helps us focus on a specific mechanism/theme instead of a general memory account as we go forward.

Similarly, the likelihood-based model comparison technique from Morton & Polyn (2016) gives us a way to evaluate the mechanism's effectiveness as part of a complete account of memory search under more general experimental conditions.

Using connected narratives rather than structured word lists to evaluate our model allows us to connect our findings, methods, and ideas to debates in the narrative comprehension literature and vice versa.

Through all this, we can validate our extension of CMR as an account not just of memory search, but of story understanding. Our toolbox of free recall analyses can begin to provide novel theoretical constraints for the study of narrative comprehension. And in turn, a vast literature of ideas and techniques exploring how rich semantic associations form can inform the broader study of semantic and episodic memory.